REAL ANALYSIS II · SEMESTER 5 · MODULE 1

Prepare: Continuous Functions

Everything you need before you open the question paper, from Plus Two and Real Analysis I up to Bartle & Sherbert §5.1, §5.3 (5.3.1–5.3.5, 5.3.7–5.3.10) and §5.4 (5.4.1–5.4.14).

File 1 of 3	**Prepare** (this file): learn the ideas
File 2	Model paper: past questions + new ones
File 3	Answer key: full solutions

Each answer in File 3 starts with topic codes such as **T8** **P7** **M4**. If a code is new to you, read that topic here first.

How to read this file

One topic at a time, in order. Every topic uses the same boxes.

REMIND ME

Facts from earlier (Plus Two, Real Analysis I) that you may have forgotten.

MATHS TOOL

A small piece of maths, taught from zero the first time it is needed.

THE IDEA

The meaning in plain words and pictures, before any symbols.

DEFINITION · SAY IT WORD FOR WORD

Bartle's exact wording. Examiners give marks for the exact words.

Underneath: what it says, in plain words.

SKELETON

The lines you can write **before** you have any idea. They earn most of the marks. ? marks the bit you must work out.

SCRATCH (ROUGH WORK, NOT FOR THE EXAM)

How we **find** the δ, the K, the point x_n or the bound. Like rough work in the margin.

PROOF (WHAT YOU WRITE)

The clean answer, in the right order. ∎

HYPOTHESES CHECKLIST

☐ Each condition of a theorem
what goes wrong without it, with a counterexample
code C2

RESULT

What you must remember and write in the exam.

EXAM TRAP

A common mistake, and how to avoid it.

CHECK YOURSELF

A question.

The answer, just below. Cover it with your thumb first.

How to study with it

> From research on how strong students learn analysis.

1. **Do, don't re-read.** Re-reading and highlighting barely help. Closing the file and writing a definition from memory helps a lot.
2. **Explain every line.** For each proof step, say *why* it is true (a definition, an earlier line, or a named theorem). The grey "why" lines show you how.
3. **Skeleton first.** In the exam, write the skeleton, leave a gap, and fill the gap last.
4. **Know the counterexamples.** Half of analysis is knowing which picture breaks a wrong idea.
5. **Little and often.** Use the recall plan on the last pages: today, in 1, 3, 7 and 14 days.

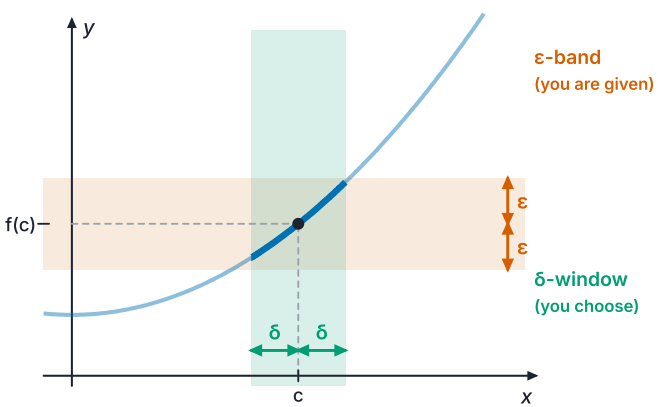
One idea for the whole module

YOU ARE HANDED E; YOU HAND BACK Δ

Continuity is a game. An opponent draws a band of half-height ε round the value f(c). You must answer with a window of half-width δ round c so that the graph over the window stays in the band.

If you can answer **every** ε, f is continuous at c. If your δ can be chosen without looking at the point c, f is **uniformly** continuous.

On a **closed bounded interval** a continuous function behaves perfectly: it is bounded, reaches its max and min, takes every value in between, and is uniformly continuous. Most of this module proves exactly that.



Continuity at c. Someone gives you a band of half-height ε round f(c). You must find a window of half-width δ round c so that the graph above the window (thick part) stays inside the band. Possible for **every** ε → continuous at c.

Codes and map

P1–P10	Plus One/Two and Real Analysis I bridge
M1–M7	Maths tools
T1–T22	Module 1 topics
C2	a counterexample from the library
Q12	question 12 of the model paper

SYLLABUS UNIT	TOPICS
1. §5.1 Continuous functions	T1–T7
2. §5.3.1–5.3.5 Bounded, max–min, roots	T8–T10
3. §5.3.7–5.3.10 Bolzano, images of intervals	T11, T12
4. §5.4 up to 5.4.3 Uniform continuity	T13–T15
5. §5.4.4–5.4.14 Lipschitz, extension, approximation	T16–T22

At least 15 of the 70 external marks. In past papers the Boundedness Theorem was asked four times, the Uniform Continuity Theorem three times, and Max–Min, Location of Roots, Preservation of Intervals and "√x is uniformly continuous" twice each.

The whole module in one story

THE IDEA

1. Continuity at a point: ε–δ, or "every sequence $x_n \to c$ gives $f(x_n) \to f(c)$". One bad sequence proves discontinuity (T1–T7).

2. On a closed bounded interval [a, b] a continuous function is bounded and attains its maximum and minimum. The proofs use Bolzano–Weierstrass (T8, T9).

3. A sign change forces a root (bisection). So a continuous function takes every value between two of its values, and maps intervals to intervals (T10–T12).

4. Uniform continuity: one δ for the whole set. Continuous on [a, b] ⇒ uniformly continuous. Lipschitz functions are uniformly continuous (T13–T19).

5. Uniform continuity on (a, b) means the ends can be filled in, and lets us approximate f by steps, broken lines and polynomials (T20–T22).

The four hypotheses to check every time

FOR THEOREMS 5.3.2, 5.3.4, 5.3.5, 5.4.3

- ☐ f is **continuous**
  a jump breaks everything **N1**
- ☐ on an **interval**
  a gap in the domain lets f skip values
- ☐ that is **closed**
  1/x on (0, 1] is unbounded **C2**
- ☐ and **bounded**
  x on [0, ∞) is unbounded; x^2 on [0, ∞) is not uniformly continuous **C1**

P1 Sets, intervals and functions

The language of every statement in this module.

REMIND ME · PLUS ONE CH 1–2

- Intervals: $[a, b]$ closed (ends included), (a, b) open, $[a, \infty)$ unbounded. "**Closed bounded interval**" means exactly $[a, b]$ with $a \leq b$ real numbers.
- A function $f : A \to \mathbb{R}$ gives one value $f(x)$ for each x in the **domain** A. The **image** of a set B is $f(B) = \{f(x) : x \in B\}$.
- Piecewise functions, the greatest integer $[x]$ (largest integer ≤ x) and signum: $\operatorname{sgn} x = 1, 0, -1$ for $x > 0$, $x = 0$, $x < 0$.

CHECK YOURSELF

Find $f([-1, 2])$ for $f(x) = x^2$.

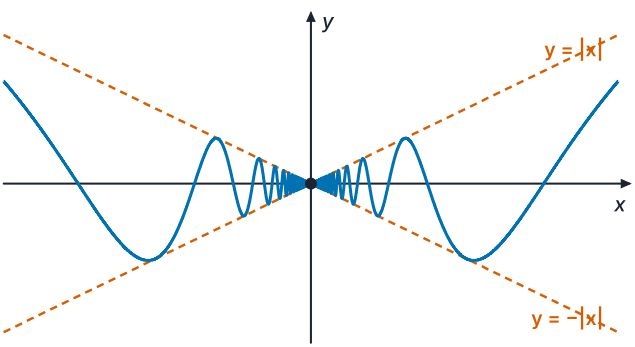
Values run from 0 (at x = 0) to 4 (at x = 2): $f([-1, 2]) = [0, 4]$. Not $[f(-1), f(2)] = [1, 4]$.

P2 Modulus (absolute value)

> $|x - c|$ is the distance between x and c.

REMIND ME · PLUS ONE

- $|x| = x$ if $x \geq 0$, $-x$ if $x < 0$. $|ab| = |a||b|$.
- $|x - c| < \delta \iff c - \delta < x < c + \delta$: "x is within δ of c".
- **Triangle inequality**: $|a + b| \leq |a| + |b|$; and $\big||a| - |b|\big| \leq |a - b|$.



x sin(1/x) (value 0 at 0) is trapped between $-|x|$ and $|x|$. So $|f(x) - 0| \leq |x| < \varepsilon$ when $|x| < \delta = \varepsilon$: **continuous at 0**, although it wiggles infinitely often.

P3 Limits and continuity in Plus Two

What you already know, now made exact.

REMIND ME · PLUS TWO CH 5

- f is continuous at c if $\lim_{x \to c} f(x) = f(c)$: left limit = right limit = value.
- Polynomials, $\sin$, $\cos$, e^x are continuous everywhere; $1/x$ is continuous wherever it is defined.
- Sums, products and quotients (non-zero denominator) of continuous functions are continuous.

WHAT CHANGES NOW

In Plus Two "limit" was a feeling ("gets close to"). Here we say exactly how close: given any ε, there is a δ. That precision lets us **prove** things like "a continuous function on [a, b] has a maximum".

P4 Rationals, irrationals and Archimedes

> Needed for Dirichlet and Thomae functions (T6).

REMIND ME · REAL ANALYSIS I, BARTLE 2.4

- **Archimedean property**: for every $\varepsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \varepsilon$.
- **Density** (2.4.8, 2.4.9): between any two real numbers there is a rational **and** an irrational. So every real number is the limit of a sequence of rationals and of a sequence of irrationals.
- Examples: $c + \sqrt{2}/n$ is irrational when c is rational, and $\to c$. Decimal truncations 1.4, 1.41, 1.414, ... are rationals $\to \sqrt{2}$.

P5 Supremum and infimum

> The maximum of f is found as a supremum (T9).

REMIND ME · BARTLE 2.3

- $\sup S$ is the least upper bound. **Completeness**: every non-empty set bounded above has a supremum in $\mathbb{R}$.
- If $s^* = \sup S$ then for each n, $s^* - \frac{1}{n}$ is not an upper bound, so some $s_n \in S$ has $s^* - \frac{1}{n} < s_n \leq s^*$.
- A supremum need not belong to S: $\sup(0,1) = 1 \notin (0,1)$.

P6 Sequences and their limit theorems

> Most proofs in §5.3 are sequence arguments.

REMIND ME · BARTLE 3.1–3.2

- $x_n \to c$: for every $\varepsilon > 0$ there is K with $|x_n - c| < \varepsilon$ for $n \geq K$.
- A convergent sequence is **bounded** (3.2.2).
- Limits of sums and products (3.2.3): $x_n \to x$, $y_n \to y$ ⇒ $x_n + y_n \to x + y$, $x_n y_n \to xy$.
- **Order** (3.2.6): if $a \leq x_n \leq b$ for all n then $a \leq \lim x_n \leq b$. Strict "<" may become "≤" in the limit.
- **Squeeze** (3.2.7): $y_n \leq x_n \leq z_n$ and $y_n, z_n \to L$ ⇒ $x_n \to L$.

P7 Subsequences and Bolzano–Weierstrass

> The engine of the Boundedness, Max–Min and Uniform Continuity Theorems.

REMIND ME · BARTLE 3.4.8

Bolzano–Weierstrass: every bounded sequence of real numbers has a convergent subsequence.

A subsequence (x_{n_r}) uses indices $n_1 < n_2 < \cdots$, so $n_r \geq r$.

WHY IT MATTERS HERE

If all x_n lie in [a, b], BW gives a subsequence converging to some x, and by the order theorem x is still in [a, b]. "Bounded" gives the subsequence; "closed" keeps its limit inside. That is why those two words appear in every big theorem of §5.3.

P8 Cauchy sequences

> Used in T19 and T20.

REMIND ME · BARTLE 3.5

(x_n) is Cauchy if for every $\varepsilon > 0$ there is H with $|x_n - x_m| < \varepsilon$ for all $m, n \geq H$. In $\mathbb{R}$: **convergent ⇔ Cauchy** (3.5.3, 3.5.5).

P9 Limits of functions

> Continuity is "limit = value", so we need limits.

REMIND ME · BARTLE 4.1

- c is a **cluster point** of A if every interval round c contains points of A other than c.
- $\lim_{x \to c} f(x) = L$: for every ε there is δ with $|f(x) - L| < \varepsilon$ when $x \in A$, $0 < |x - c| < \delta$.
- **Sequential criterion for limits** (4.1.8): $\lim_{x \to c} f = L$ iff $f(x_n) \to L$ for every sequence $x_n \to c$ with $x_n \neq c$.

P10 Nested intervals and what an interval is

> Used in the bisection proof (T10) and Preservation of Intervals (T12).

REMIND ME · BARTLE 2.5

- **Nested Intervals Property** (2.5.2): if $I_1 \supseteq I_2 \supseteq \cdots$ are closed bounded intervals, some point c lies in all of them. If their lengths → 0, there is exactly one such c.
- **Characterisation** (2.5.1): a set S with at least two points is an interval iff whenever $x < y$ are in S, everything between them is in S.

M1 Inequality gym

> The six moves inside every ε–δ proof. Practise until automatic.

MATHS TOOL

1. **Factor**: $x^2 - c^2 = (x-c)(x+c)$; $\dfrac{1}{x} - \dfrac{1}{c} = \dfrac{c-x}{xc}$; $\sqrt{x} - \sqrt{c} = \dfrac{x-c}{\sqrt{x}+\sqrt{c}}$.
2. **Bound the other factor** by first restricting $|x-c| < 1$: then $|x+c| \le |x-c| + 2|c| < 1 + 2|c|$.
3. **Keep away from 0**: if $|x-c| < c/2$ (with $c > 0$) then $x > c/2$, so $\dfrac{1}{x} < \dfrac{2}{c}$.
4. **Take the smaller**: if you need both $|x-c| < 1$ and $|x-c| < \varepsilon/5$, use $\delta = \min\{1, \varepsilon/5\}$.
5. **Reverse triangle**: $\big||x| - |c|\big| \le |x-c|$.
6. **AM–GM**: $1 + x^2 \ge 2|x|$, so $\dfrac{|x|}{1+x^2} \le \dfrac{1}{2}$.

CHECK YOURSELF

If $|x-3| < 1$, find a bound for $|x+3|$.

$2 < x < 4$, so $5 < x+3 < 7$: $|x+3| < 7$ (= 1 + 2·3).

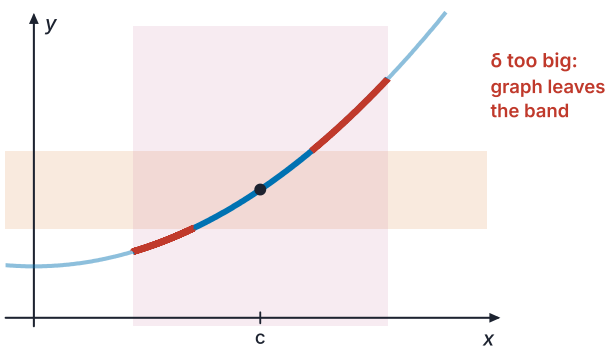
M2 The ε–δ game and its picture

How to read "for every ε there is δ".

MATHS TOOL · READ IT ALOUD

"Whatever positive ε you are handed, you can produce a positive δ such that every x of the domain within δ of c has f(x) within ε of f(c)."

PROPERTY	Δ MAY DEPEND ON
continuous at c	**ε and c**
uniformly continuous on A	**ε only**



A window that is too wide fails (red parts leave the band). That proves nothing: you are allowed to shrink δ. Continuity fails only if **no** δ works.

M3 Negating a statement

> "Not continuous", "not uniformly continuous", "not bounded": write each exactly.

MATHS TOOL · FLIP EACH WORD

"for every" ↔ "there is", and the last inequality flips.

Not continuous at c	there is $\varepsilon_0 > 0$ such that for every δ > 0 some $x \in A$ has $\lvert x - c\rvert < \delta$ but $\lvert f(x) - f(c)\rvert \geq \varepsilon_0$.
Not uniformly continuous	there is $\varepsilon_0 > 0$ such that for every δ some $x, u \in A$ have $\lvert x - u\rvert < \delta$ but $\lvert f(x) - f(u)\rvert \geq \varepsilon_0$.
Unbounded	for every $M > 0$ some $x_M \in A$ has $\lvert f(x_M)\rvert > M$.

M4 Scratch work, then proof

How to find δ, and how to write the answer.

SCRATCH · RUN BACKWARDS

Want $|f(x) - f(c)| < \varepsilon$. Factor out $|x - c|$. Bound the rest by a number B (restrict δ ≤ 1 if needed). Then $B|x - c| < \varepsilon \Leftarrow |x - c| < \varepsilon / B$.

PROOF · RUN FORWARDS

Let $\varepsilon > 0$. Choose $\delta = \min\{1, \varepsilon/B\}$. Let $x \in A$ with $|x - c| < \delta$. Then ... $\leq B|x - c| < B \cdot \dfrac{\varepsilon}{B} = \varepsilon$. ∎

EXAM TRAP

Never write a δ containing x (like $\delta = \varepsilon / |x + c|$). δ must be chosen before x is picked. Circle every symbol in your δ: only ε, c and numbers are allowed.

M5 Contradiction, and "one point for each n"

> The pattern of the three big proofs in §5.3–5.4.

MATHS TOOL · THE PATTERN

1. **Assume the opposite** (e.g. "f is not bounded").
2. **Use the negation with M = n or δ = 1/n** to get one point x_n for each $n \in \mathbb{N}$.
3. **Bolzano–Weierstrass** (P7) gives a subsequence converging to some z in [a, b].
4. **Continuity at z** (sequential form) gives $f(x_{n_r}) \to f(z)$.
5. This **contradicts** step 2. So the assumption was false.

M6 Choosing sequences to break things

> One bad sequence disproves continuity; one bad pair disproves uniform continuity.

MATHS TOOL

not continuous at c	one $x_n \to c$ with $f(x_n) \not\to f(c)$ (or two sequences giving different limits)
not uniformly continuous	two sequences x_n, u_n with $x_n - u_n \to 0$ but $\lvert f(x_n) - f(u_n) \rvert \geq \varepsilon_0$

Where to look: near a jump, near an open end where f blows up or wiggles, or far out where f gets steep.

M7 Bisection

> Halve, keep the half with the sign change, repeat.

MATHS TOOL

Start with $I_1 = [a, b]$ where f changes sign. Look at the midpoint p. If f(p) = 0, done. Otherwise keep the half where the sign still changes. After n steps the interval has length $(b - a)/2^{n-1}$, and the midpoint is within $(b - a)/2^n$ of a root.

CHECK YOURSELF

How many halvings guarantee a root of a function on [0, 1] to within 10^{-5}?

Need $1/2^n < 10^{-5}$, i.e. $2^n > 100000$: n = 17 ($2^{17} = 131072$).

Continuity at a point

T1 The ε–δ definition

> The definition everything else rests on (PYQ MTS-22 Q1).

DEFINITION 5.1.1 · SAY IT WORD FOR WORD

Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is **continuous at c** if, given any number $\varepsilon > 0$, there exists $\delta > 0$ such that if x is any point of A satisfying $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$. If f fails to be continuous at c, then we say that f is **discontinuous at c**.

In plain words: you can force f(x) as close to f(c) as anyone demands, just by taking x close enough to c.

DEFINITION 5.1.5 · CONTINUOUS ON A SET

If $B \subseteq A$, f is **continuous on B** if f is continuous at every point of B.

SKELETON · PROVE F IS CONTINUOUS AT C

Let $\varepsilon > 0$. Choose $\delta =$? . Let $x \in A$ with $|x - c| < \delta$. Then $|f(x) - f(c)| \leq$? $< \varepsilon$. Hence f is continuous at c. ∎

TWO EASIEST EXAMPLES (BARTLE 5.1.6(A), (B))

Constant $f(x) = b$: $|f(x) - f(c)| = |b - b| = 0 < \varepsilon$ for every x, so **any** δ works (say δ = 1).

Identity $g(x) = x$: $|g(x) - g(c)| = |x - c|$, so $\delta = \varepsilon$ works.

YOUR INTUITION WILL LIE TO YOU HERE

"Continuous = can be drawn without lifting the pen." The pen rule cannot prove anything, and it is wrong for $1/x$: that graph has two pieces, yet $1/x$ is continuous at every point of its domain $x \neq 0$. Continuity is only ever about points **in** the domain.

CHECK YOURSELF

For $f(x) = 3x + 1$, which δ works for a given ε?

$|f(x) - f(c)| = 3|x - c| < \varepsilon$ when $|x - c| < \varepsilon/3$. So $\delta = \varepsilon/3$ (the same at every c).

In the model paper: **Q1** (PYQ MTS-22 Q1), **Q5** .

T2 Neighbourhoods, limits and isolated points

Two other ways to say the same thing (Theorem 5.1.2).

THEOREM 5.1.2 · NEIGHBOURHOOD FORM

f is continuous at $c \in A$ iff given any ε-neighbourhood $V_\varepsilon(f(c))$ of f(c) there exists a δ-neighbourhood $V_\delta(c)$ of c such that if x is any point of $A \cap V_\delta(c)$, then f(x) belongs to $V_\varepsilon(f(c))$; that is, $f(A \cap V_\delta(c)) \subseteq V_\varepsilon(f(c))$.

$V_\delta(c) = (c-\delta, c+\delta)$. This is Definition 5.1.1 written with sets.

RESULT · LIMIT FORM (REMARKS 5.1.2)

If c is a **cluster point** of A: f is continuous at c ⇔ **(i)** f(c) is defined, **(ii)** $\lim_{x\to c} f(x)$ exists, **(iii)** the two are equal.

If c is **not** a cluster point (an isolated point of A), f is automatically continuous at c: a small window round c contains no other point of A.

CHECK YOURSELF

$A = [0,1] \cup \{2\}$, f any function on A. Is f continuous at 2?

Yes. With δ = ½ the only point of A in the window is 2 itself, so $|f(x) - f(2)| = 0 < \varepsilon$.

In the model paper: **Q2**.

T3 The sequential criterion

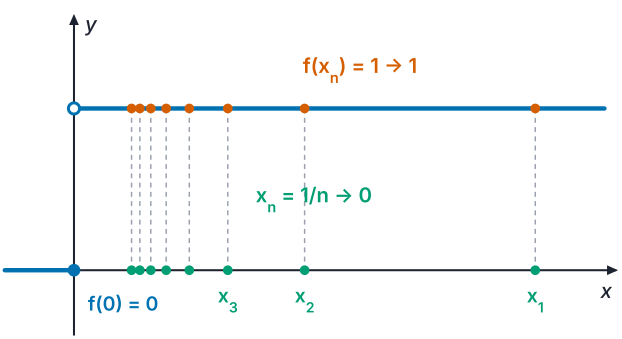
Continuity through sequences: the tool for disproving continuity (PYQ MTS-26 Q1).

THEOREM 5.1.3 · SEQUENTIAL CRITERION FOR CONTINUITY

A function $f : A \to \mathbb{R}$ is continuous at the point $c \in A$ if and only if for every sequence (x_n) in A that converges to c, the sequence $(f(x_n))$ converges to f(c).

THEOREM 5.1.4 · DISCONTINUITY CRITERION

Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. Then f is discontinuous at c if and only if there exists a sequence (x_n) in A such that (x_n) converges to c, but the sequence $(f(x_n))$ does not converge to f(c).



Sequential criterion. Points $x_n = 1/n$ march to 0, but the values $f(x_n) = 1$ stay away from $f(0) = 0$. One such sequence is enough to show f is not continuous at 0.

PROOF OF 5.1.3 (BARTLE LEAVES IT TO YOU)

(⇒)

1. Let $x_n \to c$, $x_n \in A$, and let $\varepsilon > 0$. Take δ from the continuity of f at c.
   Definition 5.1.1.

2. Since $x_n \to c$, there is K with $|x_n - c| < \delta$ for $n \geq K$.
   Definition of convergence, used with the number δ in place of ε.

3. So $|f(x_n) - f(c)| < \varepsilon$ for $n \geq K$: $f(x_n) \to f(c)$.
   Step 1 applied to $x = x_n$.

(⇐) by contrapositive

1. Suppose f is not continuous at c: some $\varepsilon_0 > 0$ has, for every δ, a point $x \in A$ with $|x - c| < \delta$, $|f(x) - f(c)| \geq \varepsilon_0$.
   Negation (M3).

2. With δ = 1/n get $x_n \in A$, $|x_n - c| < 1/n$, $|f(x_n) - f(c)| \geq \varepsilon_0$.
   One point for each n (M5).

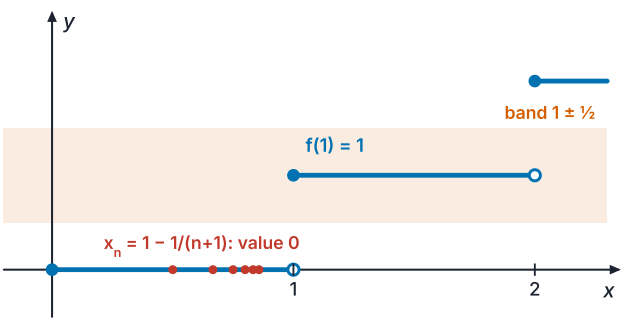
3. Then $x_n \to c$ but $f(x_n) \not\to f(c)$: the sequence condition fails. ∎
   Squeeze for $x_n \to c$; the gap stays ≥ ε_0. Steps 4–6 also prove 5.1.4.

SKELETON · PROVE F IS DISCONTINUOUS AT C

Take $x_n =$? $\in A$. Then $x_n \to c$, but $f(x_n) =$? $\to$? $\neq f(c)$. By the Discontinuity Criterion, f is discontinuous at c. ∎

WORKED EXAMPLE · GREATEST INTEGER AT 1

$f(x) = [x]$, $c = 1$, $f(1) = 1$. Take $x_n = 1 - \frac{1}{n+1} \in [\frac{1}{2}, 1)$. Then $x_n \to 1$ and $[x_n] = 0$, so $f(x_n) \to 0 \neq 1$. Discontinuous at 1.



Greatest integer f(x) = [x] at c = 1. The points $x_n = 1 - 1/(n+1)$ tend to 1, but $f(x_n) = 0$ stays outside the band 1 ± ½. By the Discontinuity Criterion, [x] is discontinuous at 1 (and at every integer).

EXAM TRAP

To **prove** continuity by sequences you need **every** sequence. To **disprove** it, **one** bad sequence is enough.

In the model paper: **Q3** (PYQ MTS-26 Q1), **Q21** .

T4 ε–δ proofs: x^2, polynomials, 1/x, |x|

Finding δ in practice (PYQ MTS-25 Q1; related FYUGP CALC-24 Q12).

Know $h(x) = x^2$, c any real number, ε > 0 given.
Want δ with $|x^2 - c^2| < \varepsilon$ whenever $|x - c| < \delta$.

SCRATCH · BARTLE 5.1.6(C)

$|x^2 - c^2| = |x - c|\,|x + c|$. Restrict $|x - c| < 1$: then $|x + c| \le |x - c| + 2|c| < 1 + 2|c|$.

Want $(1 + 2|c|)\,|x - c| < \varepsilon \Leftarrow |x - c| < \dfrac{\varepsilon}{1 + 2|c|}$.

PROOF

Let $\varepsilon > 0$ and put $\delta = \min\left\{1, \dfrac{\varepsilon}{1 + 2|c|}\right\}$. If $|x - c| < \delta$, then $|x + c| < 1 + 2|c|$, so

$$|x^2 - c^2| = |x - c||x + c| < \frac{\varepsilon}{1 + 2|c|}(1 + 2|c|) = \varepsilon.$$

So x^2 is continuous at every c. ∎

CHECK YOURSELF

Find δ for x^2 at c = 3, ε = 0.01.

$\delta = \min\{1, 0.01/7\} = 1/700 \approx 0.00143.$

RESULT · POLYNOMIALS (PYQ MTS-25 Q1)

If $x_n \to c$ then $x_n^k \to c^k$ and $a_k x_n^k + \cdots + a_0 \to a_k c^k + \cdots + a_0$ (limits of sums and products, P6). By the sequential criterion, **every polynomial is continuous on $\mathbb{R}$.**

SCRATCH · $\varphi(x) = 1/x$ AT $c > 0$ (BARTLE 5.1.6(D))

$\left|\frac{1}{x} - \frac{1}{c}\right| = \frac{|x-c|}{xc}$. Keep x away from 0: if $|x - c| < c/2$ then $x > c/2$, so $\frac{1}{xc} < \frac{2}{c^2}$. Want $\frac{2}{c^2}|x-c| < \varepsilon$.

PROOF

Given ε > 0 put $\delta = \min\left\{\frac{c}{2}, \frac{c^2\varepsilon}{2}\right\}$. If $|x-c| < \delta$ then $\left|\frac{1}{x} - \frac{1}{c}\right| < \frac{2}{c^2} \cdot \frac{c^2\varepsilon}{2} = \varepsilon$. ∎

(c = ½, ε = 1/10: δ = min{¼, 1/80} = 1/80.)

$|x|$ IS CONTINUOUS EVERYWHERE (BARTLE EX 5.1.10)

$\big||x| - |c|\big| \le |x-c|$ (reverse triangle inequality, M1). So $\delta = \varepsilon$ works at every c.

EXAM TRAP

Forgetting the "min": $\delta = \varepsilon/(1+2|c|)$ alone is wrong, because the bound $|x+c| < 1 + 2|c|$ needs $|x-c| < 1$ too.

In the model paper: **Q4** (CALC-24 Q12), **Q22** (PYQ MTS-25 Q1), **Q23** .

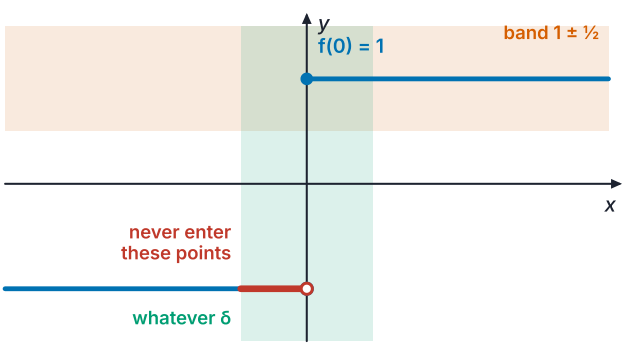
T5 Where functions break

1/x at 0, signum, jumps, joined pieces (old MCQs; related FYUGP CALC-24 Q3).

BARTLE 5.1.6(E), (F)

1/x at 0: 0 is not in the domain, so the question "continuous at 0?" does not arise; and no value at 0 can help, since $x_n = 1/n$ gives $1/x_n = n \to \infty$.

sgn at 0: $x_n = 1/n$ gives $\operatorname{sgn} x_n = 1 \not\to 0 = \operatorname{sgn} 0$. Discontinuous at 0; continuous at every $c \neq 0$ (constant near c: δ = |c|).



A jump at 0 (f = −1 for x < 0, f = 1 for x ≥ 0). Take ε = ½. Every window round 0 contains points with x < 0, where f = −1, far outside the band. So f is **not** continuous at 0.

RESULT · NAMES USED IN OLD MCQS (GENERAL KNOWLEDGE)

removable	limit exists but ≠ f(c) (or f(c) undefined): "a hole"
jump	left and right limits exist but differ (sgn at 0)
infinite	f blows up (1/x near 0)

JOINING TWO PIECES (BARTLE EX 5.1.3; CALC-24 Q3)

If f is continuous on [a, b], g on [b, c] and **f(b) = g(b)**, the joined function is continuous on [a, c]. If the values at b differ, it jumps.

Example: $f(x) = x^2$ $(x \leq 2)$, $x + 10$ $(x > 2)$: f(2) = 4, left limit 4, right limit 12, a jump of 8. Using $x + 2$ instead makes it continuous.

CHECK YOURSELF · OLD MCQ

Which of sin x, sin(1/x), 1/x are continuous functions on $A = \{x \geq 0\}$?

Only sin x: the other two are not even defined at 0 ∈ A.

EXAM TRAP

$[\sin x]$ jumps at every $k\pi$ **and** at every $\pi/2 + 2k\pi$ (where it is suddenly 1).

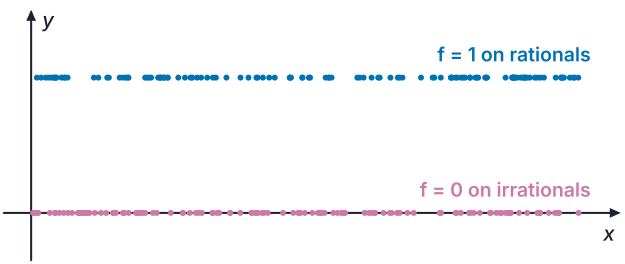
In the model paper: **Q6** (CALC-24 Q3), **Q7** (PYQ MAT-22 MCQ 3, 8; MAT-21 MCQ 6).

T6 Dirichlet and Thomae

Two famous functions you cannot draw (PYQ MTS-25 Q16; MTS-22 MCQ 1).

DIRICHLET'S FUNCTION (BARTLE 5.1.6(G))

$f(x) = 1$ if x is rational, $f(x) = 0$ if x is irrational. f is not continuous at any point of $\mathbb{R}$.



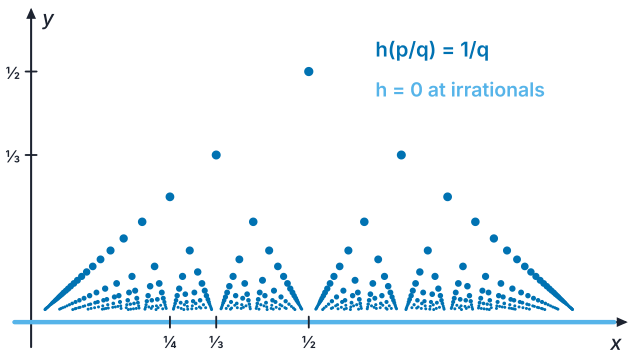
Dirichlet's function (a cartoon: both sets are dense, so the real picture is two 'full' lines). Every window round any c contains points of both kinds, so the values jump between 0 and 1: **continuous nowhere**.

PROOF

1. c rational: the irrationals $x_n = c + \sqrt{2}/n \to c$, and $f(x_n) = 0 \to 0 \neq 1 = f(c)$.
   Density of irrationals (P4).
2. b irrational: choose rationals $y_n \to b$ (e.g. decimal truncations); $f(y_n) = 1 \to 1 \neq 0 = f(b)$.
   Density of rationals.
3. By the Discontinuity Criterion f is discontinuous at every point. ∎
   Theorem 5.1.4.

THOMAE'S FUNCTION (BARTLE 5.1.6(H))

On $A = \{x : x > 0\}$: $h(x) = 0$ if x is irrational; $h(m/n) = 1/n$ if $m, n \in \mathbb{N}$ have no common factor except 1. h is continuous at every irrational in A and discontinuous at every rational in A.



Thomae's function on (0,1) (the book's h). Only finitely many dots are higher than any given ε, so near an irrational point the graph is within ε of 0: **continuous at every irrational, discontinuous at every rational**.

PROOF

1. Rational $a = m/n$: irrationals $x_k \to a$ give $h(x_k) = 0 \to 0 \neq 1/n = h(a)$. Discontinuous.
   Discontinuity Criterion.

2. Irrational b, ε > 0: choose n_0 with $1/n_0 < \varepsilon$.
   Archimedes (P4).

3. In $(b-1, b+1)$ only finitely many rationals have denominator $< n_0$.
   For each denominator q there are finitely many p with $(b-1)q < p < (b+1)q$.

4. Choose δ > 0 less than the distance from b to each of them.
   Finitely many positive distances have a smallest one; b itself is not among them.

5. If $|x - b| < \delta$: x is irrational (h = 0) or has denominator ≥ n_0 (h ≤ $1/n_0$). So $|h(x) - h(b)| < \varepsilon$. ∎
   h(b) = 0.

At $b = \sqrt{2}/2 \approx 0.7071$ with ε = 0.2: $n_0 = 6$; the nearest rationals with denominator ≤ 5 are 2/3 and 3/4 (distances 0.040, 0.043), so δ = 0.04 works.

In the model paper: **Q24** (PYQ MTS-25 Q16), **Q25** (MTS-22 MCQ 1).

T7 Filling holes: x sin(1/x) versus sin(1/x)

When can a missing value be filled in? (Bartle 5.1.7–5.1.8; MAT-22 MCQ 9.)

RESULT · REMARK 5.1.7

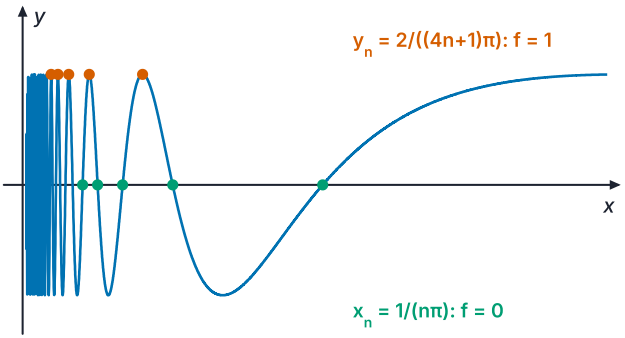
If f has a limit L at c (c not in the domain), defining F(c) = L gives a function continuous at c. If f has **no** limit at c, no value at c can make it continuous.

EXAMPLES

$\dfrac{x^2+x-6}{x-2} = x+3$ for $x \neq 2$; the limit at 2 is 5. Define f(2) = 5: continuous (a removable hole).

$F(x) = x\sin(1/x)$, F(0) = 0: $|F(x)-0| \leq |x| < \varepsilon$ when $|x| < \delta = \varepsilon$. Continuous at 0.

$\sin(1/x)$: $x_n = \dfrac{1}{n\pi}$ gives 0, $y_n = \dfrac{2}{(4n+1)\pi}$ gives 1. Two limits, so no limit: no continuous extension.



sin(1/x) wiggles faster and faster near 0. Two sequences tending to 0 give different limits (0 and 1), so no value at 0 can make it continuous there.

In the model paper: **Q8** (PYQ MAT-22 MCQ 9).

Continuous functions on intervals

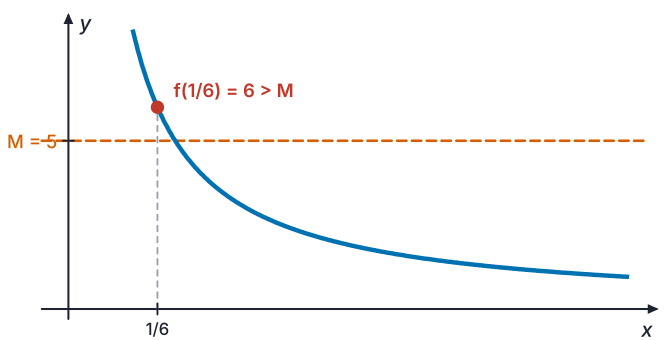
T8 Bounded functions; the Boundedness Theorem

The most-asked theorem of the module (PYQ MTS-26 Q16, MAT-22 Q27, MAT-21 Q13, MTS-22 Q2).

DEFINITION 5.3.1 · SAY IT WORD FOR WORD

A function $f : A \to \mathbb{R}$ is said to be **bounded on A** if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for all $x \in A$.

Not bounded: for every $M > 0$ there is $x_M \in A$ with $|f(x_M)| > M$.



Continuous but unbounded: f(x) = 1/x on (0, ∞). Whatever M you name, x_M = 1/(M+1) gives f(x_M) = M + 1 > M. (The domain is not closed and bounded, so the Boundedness Theorem does not apply.)

THEOREM 5.3.2 · BOUNDEDNESS THEOREM

Let $I := [a, b]$ be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f is bounded on I.

Know f continuous at every point of [a, b].

Want some M with $|f(x)| \leq M$ for all x in [a, b].

PROOF (BY CONTRADICTION, PATTERN M5)

1. Suppose f is not bounded on I. Then for each $n \in \mathbb{N}$ there is $x_n \in I$ with $|f(x_n)| > n$.
   Negation of bounded (M3), with M = n.

2. I is bounded, so (x_n) is bounded; by Bolzano–Weierstrass a subsequence (x_{n_r}) converges to some x.
   "Bounded" is used here (P7).

3. $a \leq x_{n_r} \leq b$, so $a \leq x \leq b$: $x \in I$.
   "Closed" is used here (order theorem 3.2.6).

4. f is continuous at x, so $f(x_{n_r}) \to f(x)$; a convergent sequence is bounded.
   "Continuous" is used here (T3, P6).

5. But $|f(x_{n_r})| > n_r \geq r$ for all r, so $(f(x_{n_r}))$ is unbounded. Contradiction. ∎
   $n_1 < n_2 < \cdots$ gives $n_r \geq r$.

HYPOTHESES CHECKLIST (WHAT FAILS WITHOUT EACH)

- ☐ interval bounded
  $f(x) = x$ on $[0, \infty)$: continuous, unbounded
- ☐ interval closed
  $g(x) = 1/x$ on $(0, 1]$: continuous, unbounded **C2**
- ☐ f continuous
  $h(x) = 1/x$ on (0, 1], $h(0) = 1$: on [0, 1], unbounded **N2**

In the model paper: **Q9** (PYQ MTS-26 Q2), **Q10** , **Q26** (PYQ MTS-26 Q16).

T9 The Maximum–Minimum Theorem

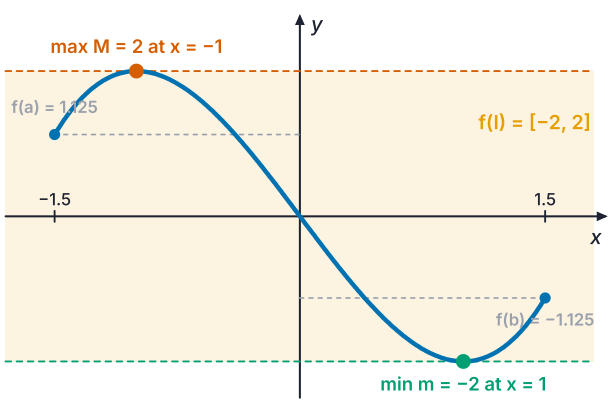
A 10–13 mark question (PYQ MAT-22 Q36a, MAT-21 Q14).

DEFINITION 5.3.3 · ABSOLUTE MAX AND MIN

f has an **absolute maximum on A** if there is a point $x^* \in A$ such that $f(x^*) \geq f(x)$ for all $x \in A$, and an **absolute minimum** if there is $x_* \in A$ with $f(x_*) \leq f(x)$ for all $x \in A$.

THEOREM 5.3.4 · MAXIMUM–MINIMUM THEOREM

Let $I := [a, b]$ be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f has an absolute maximum and an absolute minimum on I.



Maximum–minimum theorem for f(x) = x³ − 3x on I = [−1.5, 1.5]. The highest value 2 and the lowest −2 are reached at interior points. The image f(I) is the closed interval [−2, 2] (amber), **not** [f(b), f(a)] = [−1.125, 1.125].

PROOF (MAXIMUM; BARTLE)

1. $f(I)$ is non-empty and bounded (Theorem 5.3.2), so $s^* := \sup f(I)$ exists.
   Completeness (P5).

2. For each n, $s^* - \frac{1}{n}$ is not an upper bound, so there is $x_n \in I$ with $s^* - \frac{1}{n} < f(x_n) \leq s^*$.
   Property of sup (P5).

3. By Bolzano–Weierstrass, a subsequence $x_{n_r} \to x^*$, and $x^* \in I$.
   I bounded (BW), I closed (3.2.6).

4. f is continuous at x^*, so $f(x_{n_r}) \to f(x^*)$.
   Sequential criterion (T3).

5. Also $s^* - \frac{1}{n_r} < f(x_{n_r}) \leq s^*$, so $f(x_{n_r}) \to s^*$.
   Squeeze (P6).

6. Limits are unique: $f(x^*) = s^* \geq f(x)$ for all $x \in I$. So x^* is an absolute maximum point.
   s^* is an upper bound of f(I).

7. Minimum: apply steps 1–6 to $-f$ (or use inf). ∎
   A maximum of −f is a minimum of f.

HYPOTHESES CHECKLIST

☐ closed

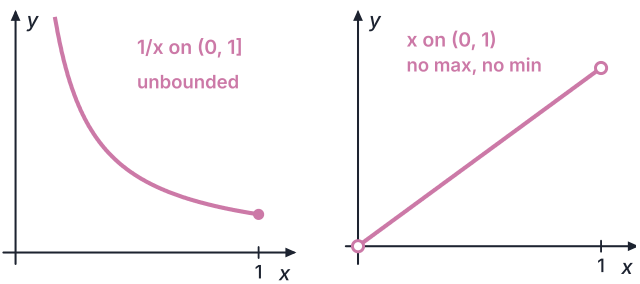
$f(x) = x$ on (0, 1): sup 1, inf 0, neither reached

☐ bounded

x on $[0, \infty)$: no maximum

☐ continuous

$h(x) = x$ $(0 \leq x < 1)$, $h(1) = 0$: sup 1 never reached



Drop "closed" or "bounded" and the theorem fails. Left: 1/x on (0, 1] is continuous but has no upper bound **C2**. Right: x on (0, 1) never reaches 0 or 1.

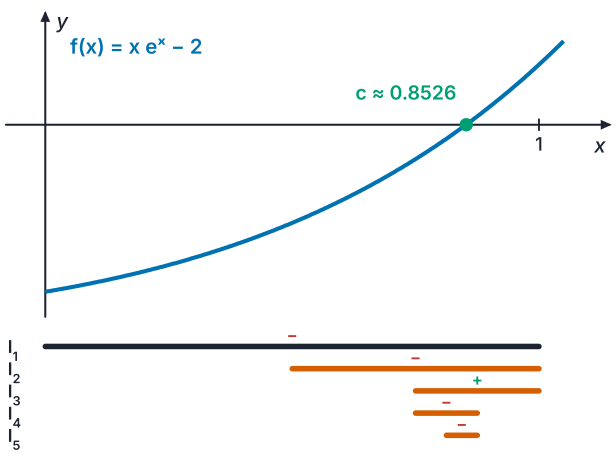
In the model paper: **Q10**, **Q11** (MAT-22 MCQ 10), **Q38** (PYQ MAT-22 Q36a, also MAT-21 Q14, MAT-22 MCQ 11).

T10 Location of Roots (bisection)

A 10-mark question (PYQ MTS-25 Q24, MAT-21 Q17).

THEOREM 5.3.5 · LOCATION OF ROOTS THEOREM

Let $I = [a, b]$ and let $f : I \to \mathbb{R}$ be continuous on I. If $f(a) < 0 < f(b)$, or if $f(a) > 0 > f(b)$, then there exists a number $c \in (a, b)$ such that $f(c) = 0$.



Bisection (proof of the Location of Roots Theorem) for f(x) = $xe^x - 2$ on [0, 1]: f(0) = −2 < 0 < e − 2 = f(1). Check the sign at each midpoint and keep the half where the sign changes. I_1 = [0, 1], I_2 = [0.5, 1], I_3 = [0.75, 1], I_4 = [0.75, 0.875], I_5 = [0.8125, 0.875] close in on c ≈ 0.8526.

PROOF (CASE F(A) < 0 < F(B))

1. $I_1 = [a, b]$, midpoint p_1. If $f(p_1) = 0$, take $c = p_1$. If $f(p_1) > 0$ let $I_2 = [a, p_1]$; if $f(p_1) < 0$ let $I_2 = [p_1, b]$.
   Keep a sign change: $f(a_2) < 0 < f(b_2)$ (M7).

2. Repeat. Either some $f(p_n) = 0$, or we get nested $I_n = [a_n, b_n]$ with $f(a_n) < 0 < f(b_n)$ and $b_n - a_n = \dfrac{b-a}{2^{n-1}}$.
   Each step halves the length.

3. By the Nested Intervals Property some c lies in every I_n; since $b_n - a_n \to 0$, $a_n \to c$ and $b_n \to c$.
   P10; $|a_n - c| \le b_n - a_n$.

4. f is continuous at c, so $f(a_n) \to f(c)$ and $f(b_n) \to f(c)$.
   Sequential criterion.

5. $f(a_n) < 0$ gives $f(c) \le 0$; $f(b_n) > 0$ gives $f(c) \ge 0$. So $f(c) = 0$.
   Order theorem: "<" becomes "≤" in the limit.

6. $c \ne a, b$ since $f(a) \ne 0 \ne f(b)$, so $c \in (a, b)$. Case $f(a) > 0 > f(b)$: apply this to $-f$. ∎
   –f changes sign the other way.

EXAM TRAP

"$f(a_n) < 0$ so $f(c) < 0$" is wrong: limits of negative numbers can be 0. Always weaken to ≤.

Using it (most common short question)

SKELETON · "SHOW THE EQUATION HAS A ROOT IN (A, B)"

Let $f(x) =$? . f is continuous on [a, b] (polynomial / exponential ...). $f(a) =$? < 0 and $f(b) =$? > 0. By the Location of Roots Theorem there is $c \in (a, b)$ with $f(c) = 0$. ∎

WORKED EXAMPLES

$xe^x - 2 = 0$ on (0, 1): $f(0) = -2 < 0 < e - 2 \approx 0.718 = f(1)$. Root in (0, 1) (it is ≈ 0.8526). (PYQ MAT-21 Q17)

$6x^3 - 3x^2 + 2$: $f(-1) = -7 < 0 < 2 = f(0)$, so a zero in (−1, 0) ⊂ (−1, 1). (Old MCQ: answer "a zero in (−1, 1)".)

$x = \cos x$: $g(x) = x - \cos x$, $g(0) = -1$, $g(\pi/2) = \pi/2$: root ≈ 0.7391.

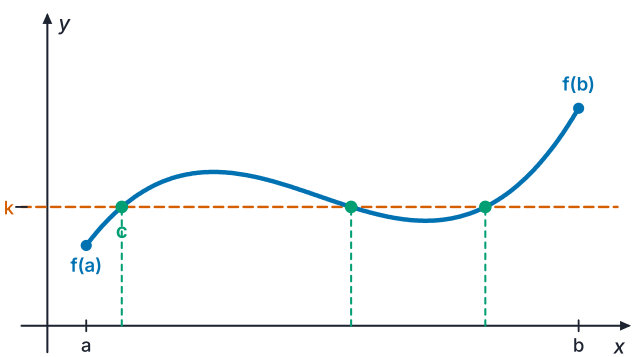
In the model paper: **Q12** (PYQ MAT-21 Q17, MAT-22 MCQ 12), **Q37** (PYQ MTS-25 Q24).

T11 Bolzano's Intermediate Value Theorem

Every value between two values is taken (PYQ MAT-21 Q1, MAT-22 Q13).

THEOREM 5.3.7 · BOLZANO'S INTERMEDIATE VALUE THEOREM

Let I be an interval and let $f : I \to \mathbb{R}$ be continuous on I. If $a, b \in I$ and if $k \in \mathbb{R}$ satisfies $f(a) < k < f(b)$, then there exists a point $c \in I$ between a and b such that $f(c) = k$.



Bolzano's intermediate value theorem. f is continuous on [a, b] and k lies between f(a) and f(b), so the graph must cross the line y = k at least once: some c has f(c) = k. (There may be several c, as here.)

PROOF

If $a < b$, let $g(x) = f(x) - k$: g is continuous and $g(a) < 0 < g(b)$, so by 5.3.5 some $c \in (a, b)$ has $g(c) = 0$, i.e. $f(c) = k$. If $b < a$, use $h(x) = k - f(x)$ on [b, a]. ∎

COROLLARY 5.3.8

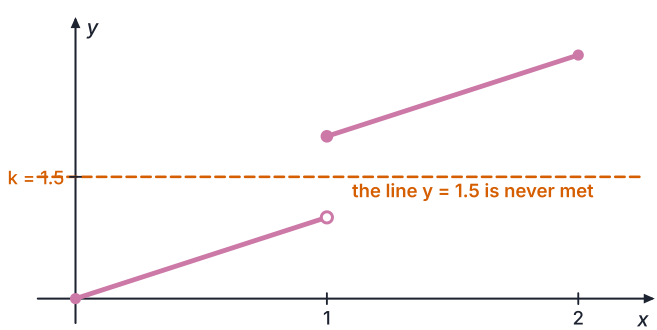
Let $I = [a, b]$ be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. If $k \in \mathbb{R}$ is any number satisfying $\inf f(I) \leq k \leq \sup f(I)$, then there exists a number $c \in I$ such that $f(c) = k$.

PROOF

By Max–Min there are $c_*, c^* \in I$ with $f(c_*) = \inf f(I)$ and $f(c^*) = \sup f(I)$. If k equals one of these, done. Otherwise $f(c_*) < k < f(c^*)$ and Bolzano (on the interval between c_* and c^*) gives c with $f(c) = k$. ∎

HYPOTHESES CHECKLIST

- ☐ f continuous
  a jump skips values **N1**
- ☐ on an interval
  $1/x$ on $\mathbb{R} \setminus \{0\}$ takes −1 and 1 but never 0



Why continuity is needed in Bolzano. f(x) = x on [0, 1), x + 1 on [1, 2]: f(0) = 0 < 1.5 < 3 = f(2), but no x gives f(x) = 1.5, because the graph jumps over it.

EXAM TRAP · THE CONVERSE IS FALSE

"Takes every intermediate value ⇒ continuous" is false: $\sin(1/x)$ with value 0 at 0 takes every value between any two of its values on [0, 1], yet is discontinuous at 0.

In the model paper: **Q13** (PYQ MAT-21 Q1), **Q27** (PYQ MAT-22 Q13).

T12 Images of intervals

5.3.9 and Preservation of Intervals (PYQ MAT-21 Q15, MTS-25 Q17, MAT-22 Q14).

THEOREM 5.3.9

Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then the set $f(I) := \{f(x) : x \in I\}$ is a closed bounded interval.

PROOF

Let $m = \inf f(I)$, $M = \sup f(I)$. By Max–Min both are values of f, so $f(I) \subseteq [m, M]$ and m, M ∈ f(I). By Corollary 5.3.8 every $k \in [m, M]$ is a value: $[m, M] \subseteq f(I)$. Hence $f(I) = [m, M]$. ∎

EXAM TRAP

$f([a, b])$ is $[m, M]$, **not** $[f(a), f(b)]$. For $x^3 - 3x$ on [−1.5, 1.5]: $[f(b), f(a)] = [-1.125, 1.125]$ but $f(I) = [-2, 2]$.

THEOREM 5.3.10 · PRESERVATION OF INTERVALS

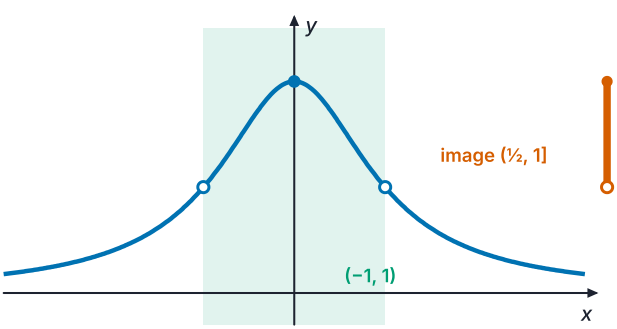
Let I be an interval and let $f : I \to \mathbb{R}$ be continuous on I. Then the set f(I) is an interval.

PROOF

1. Let $\alpha < \beta$ be in f(I): $\alpha = f(a)$, $\beta = f(b)$ for some $a, b \in I$.
   Meaning of image.

2. If $\alpha < k < \beta$, Bolzano gives $c \in I$ between a and b with $f(c) = k$, so $k \in f(I)$.
   T11; c ∈ I because I is an interval.

3. So $[\alpha, \beta] \subseteq f(I)$ whenever $\alpha < \beta$ are in f(I). By the characterisation of intervals (2.5.1), f(I) is an interval. ∎
   P10. (If f is constant, f(I) is a single point, a degenerate interval.)



f(x) = 1/(1 + x²) maps the open interval (−1, 1) onto (½, 1], which is an interval but not open. Continuous functions keep intervals as intervals (Theorem 5.3.10), but not "open" or "closed" (unless the interval is closed and bounded).

CHECK YOURSELF

Find $f([0,\infty))$ and $f([-1,1])$ for $f(x)=\dfrac{1}{x^2+1}$.

$(0,1]$ (closed interval mapped to a non-closed one: allowed, as $[0,\infty)$ is not bounded); $[\frac{1}{2},1]$.

In the model paper: **Q28** (PYQ MAT-21 Q15), **Q29** (PYQ MTS-25 Q17, also MAT-22 Q14).

One δ for the whole set

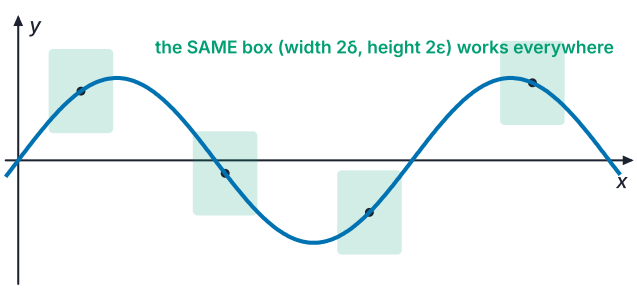
T13 Uniform continuity

The definition (PYQ MAT-22 Q1, MAT-21 Q2, old MCQs).

DEFINITION 5.4.1 · SAY IT WORD FOR WORD

Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. We say that f is **uniformly continuous on A** if for each $\varepsilon > 0$ there is a $\delta(\varepsilon) > 0$ such that if $x, u \in A$ are any numbers satisfying $|x - u| < \delta(\varepsilon)$, then $|f(x) - f(u)| < \varepsilon$.

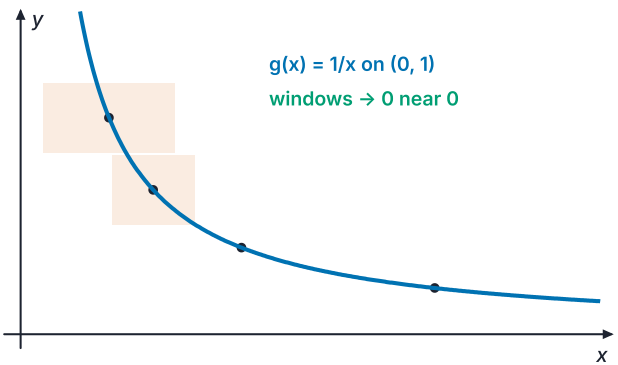
In plain words: one δ, chosen from ε alone, works at every point. Picture: a box of width 2δ and height 2ε slides along the whole graph and the curve leaves only through its sides.



sin x on $\mathbb{R}$: $|\sin x - \sin u| \le |x - u|$, so $\delta = \varepsilon$ works at every point. One rectangle slides along the whole graph and the curve leaves only through its sides. That is **uniform continuity**.

RESULT

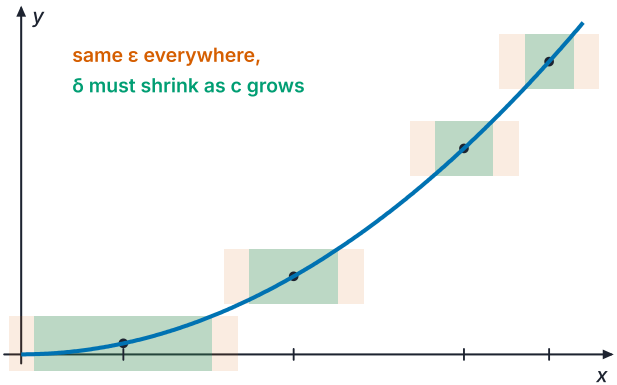
Uniformly continuous on A ⇒ continuous at every point of A (use the same δ at each point). The converse is **false**: $g(x) = 1/x$ on $(0, \infty)$ is continuous but not uniformly continuous.



g(x) = 1/x on (0, 1): a **bounded** domain, still not uniformly continuous. Near 0 the graph is so steep that δ must go to 0 C2. (Uniform failure comes from an unbounded domain OR from blowing up / wiggling at an open end.)

WHY 1/X FAILS (BARTLE'S DISCUSSION)

At a point u the best δ is about $u^2\varepsilon$ (Bartle uses $\delta = \min\{\frac{u}{2}, \frac{u^2\varepsilon}{2}\}$). As $u \to 0$ this shrinks to 0, so no single positive δ serves every u.



$f(x) = x^2$ on $[0, \infty)$ with the same ε at several points. The steeper the graph, the narrower the δ-window must be. No single δ works for all c, so x^2 is **not uniformly continuous** on $[0, \infty)$ C1.

EXAM TRAP

"Uniform continuity fails only on unbounded sets." No: 1/x and sin(1/x) on the bounded set (0, 1) are not uniformly continuous. The trouble is an open end where f blows up or wiggles.

In the model paper: **Q14** (PYQ MAT-22 Q1, also MAT-21 Q2, MAT-22 MCQ 13).

T14 Proving "not uniformly continuous"

Nonuniform Continuity Criteria 5.4.2 (PYQ MAT-21 Q16; MTS-22 MCQ 2).

5.4.2 NONUNIFORM CONTINUITY CRITERIA

Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. Then the following statements are equivalent:
(i) f is not uniformly continuous on A.

(ii) There exists an $\varepsilon_0 > 0$ such that for every $\delta > 0$ there are points x_δ, u_δ in A such that $|x_\delta - u_\delta| < \delta$ and $|f(x_\delta) - f(u_\delta)| \geq \varepsilon_0$.

(iii) There exists an $\varepsilon_0 > 0$ and two sequences (x_n) and (u_n) in A such that $\lim(x_n - u_n) = 0$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$.

SKELETON · USE (III)

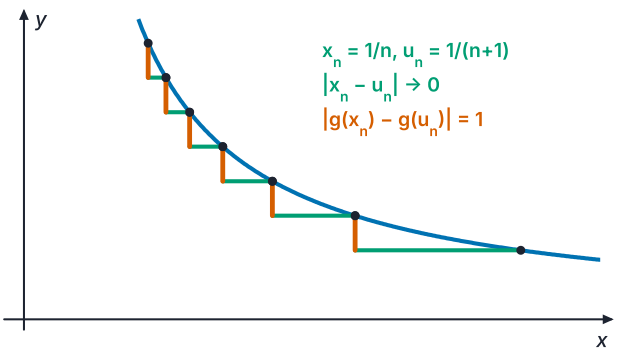
Take $\varepsilon_0 =$? , $x_n =$? , $u_n =$? in A. Then $x_n - u_n =$? $\to 0$ and $|f(x_n) - f(u_n)| =$? $\geq \varepsilon_0$ for all n. By 5.4.2, f is not uniformly continuous on A. ∎

THE STANDARD PAIRS

1/x on (0, ∞) $x_n = \frac{1}{n}$, $u_n = \frac{1}{n+1}$: gap in x is $\frac{1}{n(n+1)} \to 0$, gap in value $|n - (n+1)| = 1 = \varepsilon_0$

x² on ℝ $x_n = n$, $u_n = n + \frac{1}{n}$: values differ by $2 + \frac{1}{n^2} \geq 2$

sin(1/x) on (0, ∞) $x_n = \frac{1}{2n\pi + \pi/2}$ (value 1), $u_n = \frac{1}{2n\pi}$ (value 0): gap 1



The sequence test for "not uniformly continuous". Pairs of points get closer and closer, yet their values stay 1 apart (green = gap in x, orange = gap in value). Take $\varepsilon_0 = 1$: no δ can work.

WHY (I) ⇔ (III) (BARTLE LEAVES IT TO YOU)

(i) ⇒ (ii) is the negation of Definition 5.4.1 (M3). (ii) ⇒ (iii): take δ = 1/n to get x_n, u_n with $|x_n - u_n| < \frac{1}{n}$. (iii) ⇒ (i): if f were uniformly continuous, the δ for ε_0 would eventually exceed $|x_n - u_n|$, forcing $|f(x_n) - f(u_n)| < \varepsilon_0$: contradiction.

EXAM TRAP

One sequence is not enough here: you need a **pair** of sequences getting closer together, with values staying a fixed ε_0 apart.

In the model paper: **Q30** (PYQ MAT-21 Q16, also MTS-22 MCQ 2).

T15 The Uniform Continuity Theorem

> Asked in three papers (PYQ MTS-26 Q24a, MAT-22 Q28, MAT-21 Q27).

THEOREM 5.4.3 · UNIFORM CONTINUITY THEOREM

Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f is uniformly continuous on I.

Know f continuous at every point of I = [a, b].
Want one δ(ε) for the whole of I.

PROOF (CONTRADICTION, PATTERN M5)

1. Suppose f is not uniformly continuous on I. By 5.4.2 there are $\varepsilon_0 > 0$ and $x_n, u_n \in I$ with $|x_n - u_n| < 1/n$ and $|f(x_n) - f(u_n)| \geq \varepsilon_0$.
   Criterion (ii) with δ = 1/n.

2. I is bounded, so by BW a subsequence $x_{n_k} \to z$; I is closed, so $z \in I$.
   P7, order theorem.

3. $|u_{n_k} - z| \leq |u_{n_k} - x_{n_k}| + |x_{n_k} - z| < \frac{1}{n_k} + |x_{n_k} - z| \to 0$, so $u_{n_k} \to z$ too.
   Triangle inequality: the partners follow.

4. f is continuous at z: $f(x_{n_k}) \to f(z)$ and $f(u_{n_k}) \to f(z)$.
   Sequential criterion.

5. So $|f(x_{n_k}) - f(u_{n_k})| \to 0$, contradicting $\geq \varepsilon_0$. Hence f is uniformly continuous. ∎
   Limit of a difference.

HYPOTHESES CHECKLIST

☐ closed
1/x on (0, 1]; sin(1/x) on (0, 1] C2 C3

☐ bounded
x^2 on [0, ∞) C1

☐ continuous
a jump: no δ works across it

In the model paper: Q39 (PYQ MTS-26 Q24a, also MAT-22 Q28, MAT-21 Q27).

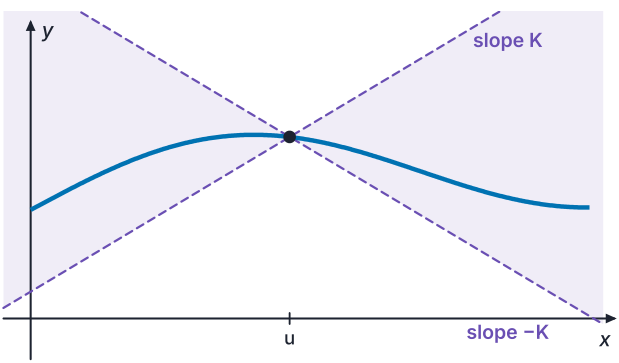
T16 Lipschitz functions

A bound on every chord slope (PYQ MTS-26 Q3, MAT-21 Q28, MTS-25 Q2, MAT-22 Q15).

DEFINITION 5.4.4 · SAY IT WORD FOR WORD

Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If there exists a constant $K > 0$ such that $|f(x) - f(u)| \leq K|x - u|$ for all $x, u \in A$, then f is said to be a **Lipschitz function** (or to satisfy a Lipschitz condition) on A.

On an interval: every chord has slope between −K and K.



Lipschitz: |f(x) − f(u)| ≤ K|x − u|. From every point u of the graph, the whole graph stays inside the double cone of slopes ±K (purple). The cone is the same shape everywhere, so δ = ε/K works for all u: **Lipschitz ⇒ uniformly continuous**.

THEOREM 5.4.5

If $f : A \to \mathbb{R}$ is a Lipschitz function, then f is uniformly continuous on A.

PROOF

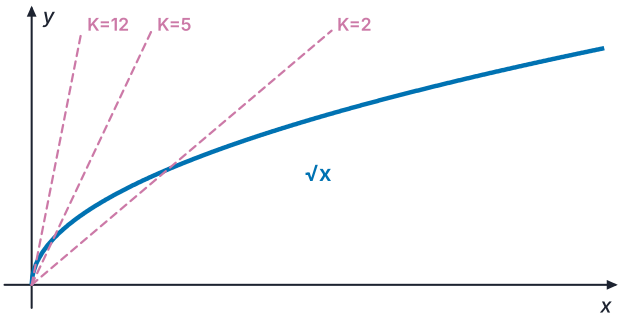
Given ε > 0 take $\delta = \varepsilon/K$. If $x, u \in A$ and $|x - u| < \delta$, then $|f(x) - f(u)| \le K|x - u| < K \cdot \frac{\varepsilon}{K} = \varepsilon$. δ uses only ε and the constant K. ∎

EXAMPLES

$3x + 1$	$\lvert f(x) - f(u)\rvert = 3\lvert x - u\rvert$: K = 3
$\sin x$ on $\mathbb{R}$	$\lvert \sin x - \sin u\rvert \le \lvert x - u\rvert$: K = 1, so δ = ε
x^2 on [0, b]	$\lvert x^2 - u^2\rvert = \lvert x + u\rvert\lvert x - u\rvert \le 2b\lvert x - u\rvert$: K = 2b
$\frac{1}{1+x^2}$ on $\mathbb{R}$	$\lvert f(x) - f(u)\rvert = \frac{\lvert u - x\rvert\lvert u + x\rvert}{(1+x^2)(1+u^2)} \le \lvert x - u\rvert$, using $\frac{\lvert x\rvert}{1+x^2} \le \frac{1}{2}$ (M1): K = 1

RESULT · UNIFORMLY CONTINUOUS BUT NOT LIPSCHITZ (PYQ MTS-25 Q2, MAT-22 Q15)

$g(x) = \sqrt{x}$ on [0, 2] is uniformly continuous (5.4.3) but not Lipschitz: for any K, the point $x = \frac{1}{(K+1)^2}$ gives $\sqrt{x} = \frac{1}{K+1} > \frac{K}{(K+1)^2} = Kx$, so $|g(x) - g(0)| > K|x - 0|$.



√x on [0, 1] is uniformly continuous (continuous on a closed bounded interval) but **not** Lipschitz: whatever K you try, near 0 the graph rises above the line y = Kx **C4**.

SUMS AND PRODUCTS OF LIPSCHITZ FUNCTIONS (OLD MCQ)

If f, g are Lipschitz with constants K_1, K_2 then $f + g$ is Lipschitz with $K_1 + K_2$, and cf with $|c|K_1$. But f^2 need not be: $f(x) = x$ on $[0, \infty)$ is Lipschitz, while x^2 has chord slopes $x + u$, unbounded. (Products are fine if f and g are also bounded.)

In the model paper: **Q15** (PYQ MTS-26 Q3, also MAT-21 Q28a), **Q16** (PYQ MTS-25 Q2, also MAT-22 Q15), **Q17** (MTS-22 MCQ 3), **Q18** (MTS-22 MCQ 4), **Q31** (PYQ MAT-21 Q28b).

T17 √x is uniformly continuous on [0, ∞)

Bartle 5.4.6(c) (PYQ MTS-26 Q24b, MAT-22 Q36b).

PROOF 1 · BARTLE'S ROUTE

1. On $I = [0, 2]$: g is continuous on a closed bounded interval, so uniformly continuous; let $\delta_I(\varepsilon)$ be its δ.
   Theorem 5.4.3.

2. On $J = [1, \infty)$: $|\sqrt{x} - \sqrt{u}| = \dfrac{|x-u|}{\sqrt{x}+\sqrt{u}} \le \dfrac{1}{2}|x-u|$. Lipschitz with $K = \dfrac{1}{2}$; $\delta_J = 2\varepsilon$.
   Conjugate (M1); $\sqrt{x} + \sqrt{u} \ge 2$ on J.

3. Put $\delta = \min\{1, \delta_I(\varepsilon), \delta_J(\varepsilon)\}$. If $|x-u| < \delta$, both points lie in I or both lie in J, so $|\sqrt{x} - \sqrt{u}| < \varepsilon$. ∎
   The "1" matters: if $x < 1 \le u$ and $|x-u| < 1$ then $u < 2$, so both are in [0, 2].

PROOF 2 · ONE LINE (GENERAL KNOWLEDGE)

For $x \ge u \ge 0$: $(\sqrt{u} + \sqrt{x-u})^2 = x + 2\sqrt{u(x-u)} \ge x$, so $\sqrt{x} - \sqrt{u} \le \sqrt{x-u}$. Hence $|\sqrt{x} - \sqrt{u}| \le \sqrt{|x-u|}$ and $\delta = \varepsilon^2$ works on all of $[0, \infty)$. ∎

In the model paper: **Q32** (PYQ MTS-26 Q24b, also MAT-22 Q36b).

T18 Sums and products of uniformly continuous functions

PYQ MAT-21 Q22.

RESULT (BARTLE EXERCISES 5.4.5–5.4.7)

If f, g are uniformly continuous on A, so is $f+g$ (take $\delta = \min\{\delta_f(\varepsilon/2), \delta_g(\varepsilon/2)\}$). The product need not be; it is if f and g are also bounded on A.

COUNTEREXAMPLE

$f(x) = g(x) = x$ on $\mathbb{R}$ are uniformly continuous (Lipschitz, K = 1). Their product x^2 is not: $x_n = n$, $u_n = n + \dfrac{1}{n}$ have $x_n - u_n \to 0$ but $|x_n^2 - u_n^2| = 2 + \dfrac{1}{n^2} \geq 2$. Another: $x \sin x$ N4.

In the model paper: Q33 (PYQ MAT-21 Q22).

T19 Uniform continuity keeps Cauchy sequences Cauchy

Theorem 5.4.7 (PYQ MAT-21 Q36a, MTS-22 Q3).

THEOREM 5.4.7

If $f : A \to \mathbb{R}$ is uniformly continuous on a subset A of $\mathbb{R}$ and if (x_n) is a Cauchy sequence in A, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.

PROOF

1. Let ε > 0 and take δ = δ(ε) from uniform continuity.
   One δ for every pair of points.
2. (x_n) is Cauchy: there is H with $|x_n - x_m| < \delta$ for $n, m > H$.
   Definition of Cauchy, used with δ (P8).
3. So $|f(x_n) - f(x_m)| < \varepsilon$ for $n, m > H$: $(f(x_n))$ is Cauchy. ∎
   Step 1 with $x = x_n$, $u = x_m$.

MERE CONTINUITY IS NOT ENOUGH

$f(x) = 1/x$ on (0, 1) is continuous; $x_n = \dfrac{1}{n+1}$ is Cauchy, but $f(x_n) = n + 1$ is unbounded, not Cauchy. (This also proves 1/x is not uniformly continuous on (0, 1).)

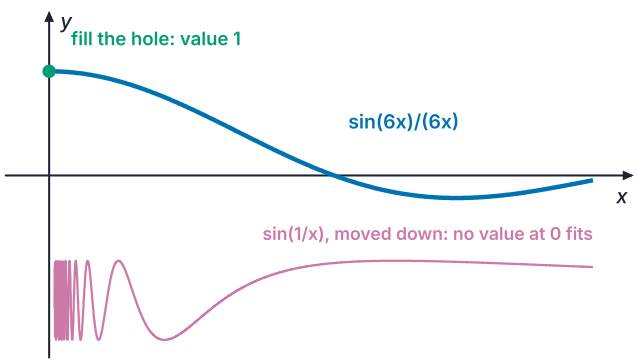
In the model paper: **Q34** (PYQ MAT-21 Q36a, also MTS-22 Q3).

T20 The Continuous Extension Theorem

A 7–10 mark question (PYQ MAT-22 Q29).

THEOREM 5.4.8 · CONTINUOUS EXTENSION THEOREM

A function f is uniformly continuous on the interval (a, b) if and only if it can be defined at the endpoints a and b such that the extended function is continuous on [a, b].



Continuous extension theorem. f is uniformly continuous on (a, b) exactly when it can be given end values that make it continuous on [a, b]. sin(6x)/(6x) → 1 at 0, so it extends; sin(1/x) has no limit at 0, so it does not, and it is not uniformly continuous on (0, 1) C3.

PROOF

(⇐) The extension is continuous on the closed bounded [a, b], hence uniformly continuous there (5.4.3), hence on the smaller set (a, b) (the same δ works).

(⇒) We define f(a); f(b) is similar.

1. Let $x_n \in (a, b)$, $x_n \to a$. It converges, so it is Cauchy.
   P8.

2. By Theorem 5.4.7, $(f(x_n))$ is Cauchy, so it converges to some L.
   Uniform continuity is used here; Cauchy ⇒ convergent (3.5.5).

3. L does not depend on the sequence: if also $u_n \to a$, then $u_n - x_n \to 0$, so by uniform continuity $f(u_n) - f(x_n) \to 0$, and $f(u_n) \to L$.
   Pair argument (M6).

4. By the sequential criterion for limits, $\lim_{x \to a} f(x) = L$. Define $f(a) := L$; f is then continuous at a.
   P9, Remark 5.1.7 (T7).

5. Likewise at b. The extended f is continuous on [a, b]. ∎
   Interior points were already fine.

USING IT

sin(1/x) has no limit at 0, so it is **not** uniformly continuous on (0, b] C3. $x \sin(1/x)$ has limit 0 at 0, so it **is** uniformly continuous on (0, b].

In the model paper: **Q40** (PYQ MAT-22 Q29).

T21 Step functions and piecewise linear approximation

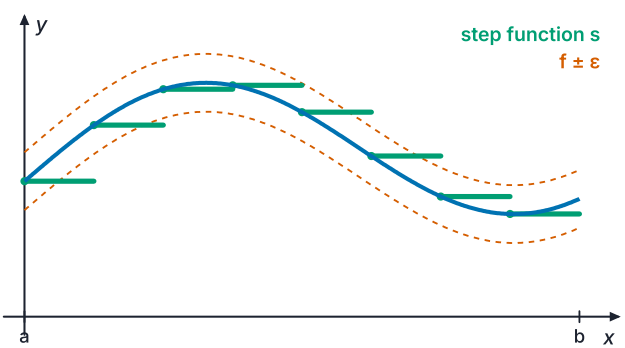
Bartle 5.4.9–5.4.13 (PYQ MAT-22 Q5, MAT-21 Q3, MTS-26 Q17).

DEFINITION 5.4.9 · STEP FUNCTION

A function $s : [a, b] \to \mathbb{R}$ is called a **step function** if [a, b] is the union of a finite number of nonoverlapping intervals $I_1, I_2, \ldots, I_n$ such that s is constant on each interval I_k.

THEOREM 5.4.10

Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. If ε > 0, then there exists a step function $s_\varepsilon : I \to \mathbb{R}$ such that $|f(x) - s_\varepsilon(x)| < \varepsilon$ for all $x \in I$.



Approximation by step functions. Because f is uniformly continuous on [a, b], cutting [a, b] into pieces shorter than δ keeps f within ε of its value at the left end of each piece. So the step function s satisfies |f(x) − s(x)| < ε for all x.

PROOF

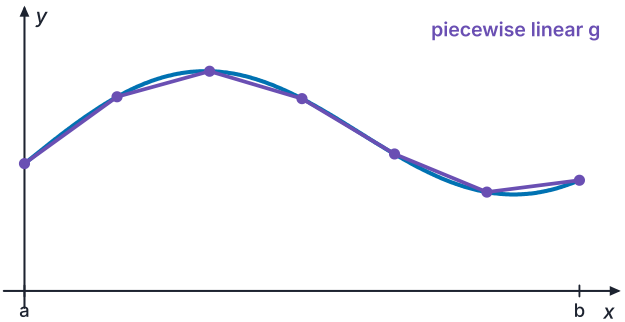
1. f is uniformly continuous on I = [a, b]: take δ(ε).
   Theorem 5.4.3: the only place continuity enters.
2. Choose m with $h = (b-a)/m < \delta$. Cut I into $I_1 = [a, a+h]$, $I_k = (a+(k-1)h, a+kh]$.
   Archimedes; the pieces do not overlap.
3. Let $s_\varepsilon(x) = f(a+kh)$ for $x \in I_k$: a step function.
   Constant on each piece.
4. For $x \in I_k$, $|x-(a+kh)| \le h < \delta$, so $|f(x) - s_\varepsilon(x)| < \varepsilon$. ∎
   Uniform continuity.

EXAM TRAP

h must be **strictly** less than δ. For f(x) = x on [0, 1], ε = 0.1 (so δ = 0.1): m = 10 gives h = 0.1 and fails at x = 0; m = 11 works (error ≤ 1/11).

DEFINITION 5.4.12 AND THEOREM 5.4.13 · PIECEWISE LINEAR

g is **piecewise linear** on [a, b] if [a, b] is a union of finitely many disjoint intervals on each of which g is linear. If f is continuous on a closed bounded I and ε > 0, there is a continuous piecewise linear g_ε with $|f(x) - g_\varepsilon(x)| < \varepsilon$ for all $x \in I$.



Approximation by a piecewise linear function: join the points $(x_k, f(x_k))$ by straight pieces. With the pieces shorter than δ, $|f - g| < \varepsilon$ everywhere. (Weierstrass: even a single polynomial can do this.)

PROOF IDEA (THE STEP BARTLE SKIPS)

Join the points $(x_k, f(x_k))$, $x_k = a + kh$, by chords. For x between x_{k-1} and x_k, write $x = (1-\lambda)x_{k-1} + \lambda x_k$; then $g(x) = (1-\lambda)f(x_{k-1}) + \lambda f(x_k)$ and $f(x) - g(x) = (1-\lambda)[f(x) - f(x_{k-1})] + \lambda[f(x) -$

where both brackets are less than ε in size. So $|f(x) - g(x)| < \varepsilon$.

CHECK YOURSELF

$f(x) = x^2$ on [0, 1] with nodes 0, ¼, ½, ¾, 1. Largest error of the broken line?

On each piece of width h the chord of x^2 lies above it by at most $h^2/4$ (at the midpoint): $\frac{1}{64} \approx 0.0156$.

In the model paper: **Q19** (PYQ MAT-22 Q5, also MAT-21 Q3), **Q35** (PYQ MTS-26 Q17), **Q36** .

T22 Weierstrass Approximation Theorem

Statement only (PYQ MTS-25 Q3, MAT-22 Q2).

THEOREM 5.4.14 · SAY IT WORD FOR WORD

Let $I = [a, b]$ and let $f : I \to \mathbb{R}$ be a continuous function. If $\varepsilon > 0$ is given, then there exists a polynomial function p_ε such that $|f(x) - p_\varepsilon(x)| < \varepsilon$ for all $x \in I$.

The proof is optional in the syllabus (Bartle omits it).

WHY "CLOSED BOUNDED" IS NEEDED

$1/x$ on (0, 1] cannot be approximated within 1 by a polynomial p: p is bounded on [0, 1], and $|1/x - p| < 1$ would make 1/x bounded too.

In the model paper: **Q20** (PYQ MTS-25 Q3, also MAT-22 Q2).

Counterexample library

> Each one breaks a tempting wrong idea. Learn the function, the set and the reason.

CODE	FUNCTION, SET	WHAT IT SHOWS
C1	x^2 on $\mathbb{R}$	continuous, unbounded, not uniformly continuous, not Lipschitz
C2	$1/x$ on (0, 1)	continuous, unbounded, no max or min, not uniformly continuous
C3	$\sin(1/x)$ on (0, 1)	bounded and continuous, not uniformly continuous, no continuous extension
C4	$\sqrt{x}$ on [0, 1]	uniformly continuous, not Lipschitz
C7	Dirichlet	continuous nowhere
C8	Thomae	continuous exactly at the irrationals
C9	sgn	jump at 0; skips the values between −1 and 1
C10	$\sin(1/x)$, value 0 at 0	takes all intermediate values, yet not continuous
C11	x on $\mathbb{Q}$, 0 off $\mathbb{Q}$	continuous only at 0
N1	x on [0, 1), $x+1$ on [1, 2]	Bolzano fails without continuity
N2	$1/x$ on (0, 1], value 1 at 0	on [0, 1] but discontinuous: unbounded
N3	x on (0, 1)	bounded, continuous, no max, no min

N4	$x \sin x$ on $\mathbb{R}$	product of uniformly continuous functions, not uniformly continuous

YOUR INTUITION WILL LIE TO YOU HERE

"A continuous function on a bounded set is bounded." C2 says no: you also need the set to be **closed**.

Which tool?

> Mixed drill. Name the tool first, then check its conditions. Answers below.

1. Show f is continuous at c.
2. Show f is not continuous at c.
3. Show an equation has a root in (a, b).
4. Show f takes the value k somewhere.
5. Show f is bounded / has a maximum.
6. Show f is uniformly continuous on [a, b].
7. Show f is uniformly continuous on an unbounded set.
8. Show f is not uniformly continuous.
9. Show f is uniformly continuous on (a, b).

ANSWERS

1. ε–δ with scratch work (T4), or "built from continuous functions". 2. One bad sequence (Discontinuity Criterion, T3). 3. Sign change + Location of Roots (T10). 4. Bolzano (T11) or Corollary 5.3.8. 5. Boundedness / Max–Min, after checking closed, bounded, continuous (T8, T9). 6. Uniform Continuity Theorem (T15). 7. Lipschitz bound (T16), or split the set (T17). 8. Pair of sequences (5.4.2, T14). 9. Continuous extension: does f have limits at a and b? (T20).

Theorem sheet

5.1.1 CONTINUITY AT C

∀ε > 0 ∃δ > 0: $x \in A$, $|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon$.

5.1.3 / 5.1.4 SEQUENCES

continuous at c ⇔ every $x_n \to c$ gives $f(x_n) \to f(c)$; discontinuous ⇔ some $x_n \to c$ with $f(x_n) \not\to f(c)$.

5.3.2 BOUNDEDNESS · 5.3.4 MAX–MIN

f continuous on closed bounded [a, b] ⇒ f bounded, and has an absolute max and min.

5.3.5 LOCATION OF ROOTS · 5.3.7 BOLZANO

f continuous on [a, b], f(a) < 0 < f(b) ⇒ f(c) = 0 for some c ∈ (a, b). f continuous on an interval, f(a) < k < f(b) ⇒ f(c) = k for some c between a and b.

5.3.9 · 5.3.10

f([a, b]) = [m, M]; f(interval) is an interval.

5.4.1 UNIFORM CONTINUITY · 5.4.3 UC THEOREM

∀ε ∃δ(ε): $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$. Continuous on [a, b] ⇒ uniformly continuous.

5.4.5 · 5.4.7 · 5.4.8

Lipschitz ⇒ UC. UC maps Cauchy sequences to Cauchy sequences. UC on (a, b) ⇔ continuously extendable to [a, b].

5.4.10 · 5.4.13 · 5.4.14

Continuous f on [a, b] is within ε of a step function, a piecewise linear function, a polynomial.

Recall plan

> Testing yourself beats re-reading. Tick each box.

DAY 0 (TODAY)

☐ Read T1–T22 with a pencil. Cover each proof and explain every "why" line aloud.

DAY 1

☐ From memory, write Definitions 5.1.1, 5.3.1, 5.4.1, 5.4.4. Check them word for word.

DAY 3

☐ Without looking, write the Boundedness Theorem proof. ☐ Find δ for x^2 at c = 2, ε = 0.1.

DAY 7

☐ Do the "Which tool?" drill. ☐ Write the Uniform Continuity Theorem and Location of Roots proofs.

DAY 14 / EXAM WEEK

☐ Say every counterexample aloud with its reason. ☐ Do the model paper, timed.

How to attack a question

3 marks	Exact definition or statement, or one example with a one-line check. About 4 minutes.
6 marks	Statement, then Know/Want, then the skeleton with the δ (or the sequences) filled in. 8–12 lines.
10 marks	Statement word for word; numbered proof with the theorem names (Bolzano–Weierstrass, sequential criterion, nested intervals); end with where each hypothesis (closed, bounded, continuous) was used. If stuck, write the skeleton anyway.

FIVE-MINUTE FINAL CHECK

Did I use every hypothesis? Is my δ free of x? Did "<" become "≤" when I took a limit? Did I say "closed and bounded"?

Past questions → topics

MTS-25 = MTS 6B 10, March 2025 · MTS-26 = April 2026 · MTS-22 = March 2022 (and its MCQ paper) · MAT-22 = MAT 6B 09, March 2022 (and MCQ) · MAT-21 = MAT 6B 09, March 2021 (and MCQ). Old CBCSS papers, same textbook. CALC-24 = FYUGP Calculus (MAT1MN102), Nov 2024: related, not this course.

PAST QUESTION	TOPIC	MODEL Q
MTS-22 Q1 define continuity; constant	T1	Q1
MTS-26 Q1 Discontinuity Criterion	T3	Q3
CALC-24 Q12 |x| continuous	T4	Q4
CALC-24 Q3 piecewise at 2	T5	Q6
MAT-22 MCQ 3, 8; MAT-21 MCQ 6	T5	Q7
MAT-22 MCQ 9 removable	T7	Q8
MTS-26 Q2; MAT-22 MCQ 6 unbounded	T8	Q9
MTS-22 Q2b, MAT-21 Q13b, Q14b examples	T8, T9	Q10
MAT-22 MCQ 10 absolute max	T9	Q11
MAT-21 Q17b; MAT-22 MCQ 12; MAT-21 MCQ 7	T10	Q12
MAT-21 Q1 Bolzano	T11	Q13
MAT-22 Q1; MAT-21 Q2; MCQs define UC	T13	Q14
MTS-26 Q3; MAT-21 Q28a Lipschitz	T16	Q15
MTS-25 Q2; MAT-22 Q15 UC not Lipschitz	T16	Q16
MTS-22 MCQ 3, 4	T16	Q17, Q18

MAT-22 Q5; MAT-21 Q3 step function	T21	Q19
MTS-25 Q3; MAT-22 Q2 Weierstrass	T22	Q20

PAST QUESTION	TOPIC	MODEL Q
MTS-25 Q1 polynomials	T4	Q22
MTS-25 Q16 Dirichlet	T6	Q24
MTS-22 MCQ 1 Thomae	T6	Q25
MTS-26 Q16; MAT-22 Q27; MAT-21 Q13a; MTS-22 Q2a Boundedness	T8	Q26
MAT-22 Q13 inf ≤ k ≤ sup	T11	Q27
MAT-21 Q15 image is [m, M]	T12	Q28
MTS-25 Q17; MAT-22 Q14 Preservation	T12	Q29
MAT-21 Q16; MTS-22 MCQ 2 nonuniform	T14	Q30
MAT-21 Q28b Lipschitz ⇒ UC	T16	Q31
MTS-26 Q24b; MAT-22 Q36b √x	T17	Q32
MAT-21 Q22 product not UC	T18	Q33
MAT-21 Q36a; MTS-22 Q3 Cauchy	T19	Q34
MTS-26 Q17 step approximation	T21	Q35
MTS-25 Q24; MAT-21 Q17a Location of Roots	T10	Q37
MAT-22 Q36a; MAT-21 Q14a; MAT-22 MCQ 11 Max–Min	T9	Q38
MTS-26 Q24a; MAT-22 Q28; MAT-21 Q27 UC Theorem	T15	Q39
MAT-22 Q29 Continuous Extension	T20	Q40